Gaia Rueda Moreno

(she/her/hers)

Urban Evolutionary Ecologist

I am a Venezuelan-American transgender woman, who is deeply curious about the natural world. My curiosity has driven me to pursue a career as a research scientist, and I am interested in the ways in which invasive species contribvute to the formation of novel urban ecosystems, and how to apply restoration ecology to an the complex socio-ecological system that is the urban jungle. I strive to share my knowledge with the world, through inclusive science communication.

Contact Information:
car9715@nyu.edu
+1 (832) 441-4723 1076 Pugsley Ave, 10472 Bronx, NY
ORCID: 0009-0001-8309-7455
iNaturalist: gaia_papaya
Github: gaia-papaya

Education:

New York University: B.A. in Biology & Environmental Studies (May 2024)

Research Experience:

REU at the Mystic Aquarium/University of Connecticut: (Summer 2022)

- Developed and executed a research project studying how food availability impacts the thermal tolerance of the estuarine copepod, ***Acartia tonsa***
- Presented my results both as a poster to a general audience, and as a talk to a scientific audience

Undergraduate Researcher at NYU Dept. of Biology: (2022-2024)

Dr. Kristin Winchell's Urban Evolutionary Ecology Lab:

- Collaborated with student colleagues to design and execute research projects, described below
- Attended meetings to discuss and present active research, and participate in professional development workshops

Project 1: Spotted Lanternfly Wing Morphology

- Collected field samples of the invasive spotted lanternfly (***L. delicatula***) throughout New York City and Shanghai
- Sampled forest invertebrate diversity in NYC Parks, to evaluate the effects of SLF invasion
- Gathered wing morphology data, including wing morphometrics and coloration, through photography and spectrometry
- Developed morphometric and geospatial analyses using R and QGIS to evaluate morphological variability, and impact of invasion on invertebrate diversity

Project 2: Luxury Effect

- Surveyed NYC parks for bird diversity, leading teams of volunteer researchers in collecting bird and environmental data
- Data-mined online databases for socio-economic and environmental data, actively developing analytical workflow for publication

Independent Research:

- Assisted a colleague working to pioneer biodegradable alternatives to nylon mesh bags used to collect aquatic invertebrates
- Provided an R based analytical workflow to analyze the effects of bag material on species abundance matrices using both traditional biodiversity indices and NMDS

Grants & Awards:

- NSF Research experience for Undergraduates (REU) fellowship (awarded summer 2022)
- NYU Dean's undergraduate Research Fund (DURF) (awarded fall 2022, spring 2023, fall 2023)
- Point BIPOC Scholar (awarded fall 2022)
- NYU Global Urban Showcase Winner (2023)

Work Experience:

Principles of Biology Peer Facilitator at NYU's Undergraduate Learning Center: (Fall 2023 - Spring 2024)

- Tutored undergraduate students in an introductory biology course, leading small-group and individual sessions
- Developed practice questions and slides to help students prepare for exams and understand biological principles

PLACE Mentor at The Bronx is Blooming: (May-August 2024)

- led a cohort of SYEP employees tasked with performing maintaining Soundview Park, in the Bronx
- coordinated daily activities, which include invasive species management, planting of native flora, and outreach with local community
- created lesson plans and educated cohort in basic ecological principles, and topics relating to environmental justice
- designed science communication materials (invasive species flashcards) to help educate mentors and employees alike on invasive plant species

Seasonal Tree Care Coordinator at the Bronx is Blooming: (Sept 2024-Ongoing)

- Led interns and volunteer groups in park stewardship efforts at Soundview park, including invasive species removal, native planting, watering and mulching
- developed and conducted citizen science initiatives, in collaboration with other staff, collecting air quality and bird biodiversity
- created maps for use in coordinating events, and for science communication purposes

Service & Outreach:

Founder & President of the Urban Naturalist Club at NYU: (Fall 2023 - Spring 2024)

- student organization dedicated to discovering the flora and fauna of New York City
- Led bird walks through various NYC parks, in all seasons, providing bird identification to attendees
- Organized social events for students interested in urban wildlife

Freelance Journalist at 'Life in The City' Urban Eco-Evo Blog: (Spring 2023 - Ongoing)

- written numerous articles highlighting novel research in urban evolutionary ecology, interesting wildlife observations, exposed the hidden connections between human and ecosystemic struggles in the urban jungle
- (https://urbanevolution-litc.com/)

Skills:

Languages:
- Spanish (native)
- English (fluent)

Computational Skills:
Coding Languages:
- R
- Python
- Shell scripting
-Javascript
- GIS software/ analyses (QGIS/QField)
- OfficeSuite
- ImageJ
- Scribus (InDesign equivalent)
- Photo editing (Darktable)
- Video/audio editing

Laboratory Skills:
- Micro-pipetting
- Microscopy Imaging
-Dynamic thermal tolerance assays
- Spectrometry
- Insect pinning & preservation

Field Skills:
- Extensive hiking experience
- Horticultural techniques (planting/removal)
-Wildlife photography
-Insect sampling (sweep netting, beating sheet, tullgren funnels)
-aquatic invertebrate sampling (leaf packs, plankton tow netting)
- Environmental data collection (Air Quality, Temp/ Humidity)

Publications:

Rueda Moreno G, Sasaki MC. *Starvation reduces thermal limits of the widespread copepod Acartia tonsa.* Ecol Evol. 2023 Oct 3;13(10):e10586. doi: 10.1002/ece3.10586. PMID: 37799447; PMCID: PMC10547671.

Morrison et al. (**Rueda Moreno Included**) *Bird Name-a-thon: Categorizing English bird names using crowdsourcing* Pre-print, Biorxiv 2024 Sep 16. https://doi.org/10.1101/2024.09.13.612884

R.E. Baez, **G. Rueda Moreno**, R. Ward, K. Schneider Paolantonio *Testing the effectiveness of biodegradable alternatives to plastic mesh bags conventionally used in freshwater macroinvertebrate analysis within the Bronx River, NYC* In preparation

G. Rueda Moreno, A. Zhang, K. Winchell *Phenotypic Differentiation of Spotted Lanternflies Lycorma delicatula in Native and Invasive Ranges*, In Preparation

Presentations:

2022:
- '*The effects of diet on the thermal limits of Acartia tonsa*' Slideshow, University of Connecticut, Avery Point Campus, CT
- '*F is For Fire and ... Food?*' Poster, Mystic Aquarium Student Research Symposium, Mystic, CT

2023:
- '*Characterizing sexual dimoprhism of L. delicatula wing morphology and color*' Slideshow, NYU, New York City NY

2024:
- '*Creepy Crawlies in the Urban Jungle: Patterns of forest Arthropod Diversity Across Urbanization Gradients*' Poster. Society for Integrated and Comparative Biology, Seattle WA
- '*DURF Interest Panel to Sophomore Scholars Program*'. NYU, New York City, NY
- '*Blending in to the Concrete Jungle: Variations in Lycorma delicatula Wing Morphology and Coloration Across Urban and Invasion Gradients*' Poster, Undergraduate Research Conference, NYU. New York City, NY